

MASA03 Lab 1

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September 2025

1 Introduction

In this lab we were tasked to explore the basics of generating random variables given their cumulative distribution functions and the ability to generate values according to continuous uniform distribution. This was done for discrete random variables and for continuous random variables.

2 Discrete random variables

Given a way to obtain a continuous random variable $X \sim \text{Un}(0,1)$, we can generate a discrete random variable with uniform distribution among $\{1, \dots, n\}$ by partitioning the unit interval into n equal sub-intervals $I_k = (k/n, (k+1)/n)$. Then, the index of the interval which X is will be the output of our discrete random variable. An efficient way to implement this is using the floor function as follows:

$$k = \lfloor nX \rfloor + 1.$$

Example 1. *The described method can be used to simulate a fair dice roll by setting $n = 6$. Then, the result will be $\lfloor 6X \rfloor + 1$.*

If we want a discrete random variable with a non-uniform distribution we will have to do more, as the unit interval will have to be partitioned into sub-intervals of different lengths corresponding to the various elementary probabilities for the discrete random variable.

Example 2. *If we take an unfair six-sided die, say, such that the probabilities are as follows:*

$$P(1) = 0.1 \quad P(2) = 0.3 \quad P(3) = 0.2 \quad P(4) = 0.05 \quad P(5) = 0.15 \quad P(6) = 0.2,$$

*then we can create an array of interval lower bounds, like this:
(0, 0.1, 0.4, 0.6, 0.65, 0.8). Then we can create a function that finds the index of the largest entry that X is larger than, which will be our dice-roll result.*

3 Continuous random variables

In the continuous case, the method for generating random variables with a specific distribution is somewhat more straightforward. Given a desired cumulative distribution function F , we can generate an $X \sim \text{Un}(0, 1)$, and then let $Y = F^{-1}(X)$. As was proven in a lecture, this will give Y the CDF F .

Example 3. Suppose one wishes to generate a random variable with the CDF $F(x) = 1 - e^{-x}$. Then $F^{-1}(x) = -\ln(1 - x)$, and thus our desired random variable, Y , can be generated with $Y = -\ln(1 - X)$ where $X \sim \text{Un}(0, 1)$. Using R we can generate a list of values distributed in this way as follows:

```
y <- -log(1-runif(1000))
```

We can then visualize these with a histogram:

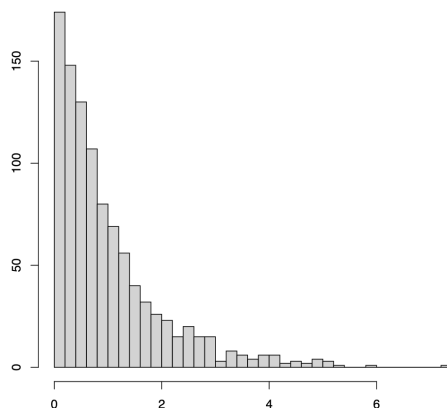


Figure 1: Exponential distribution, CDF $1 - e^{-x}$

This is of similar shape to the derivative of F , which is as expected.

It should be noted that finding F^{-1} may not be trivial. For example if we wish to generate a random variable with normal distribution the CDF does not even have an elementary form. In these cases there are numerical algorithms to approximate an inverse.

A Histograms

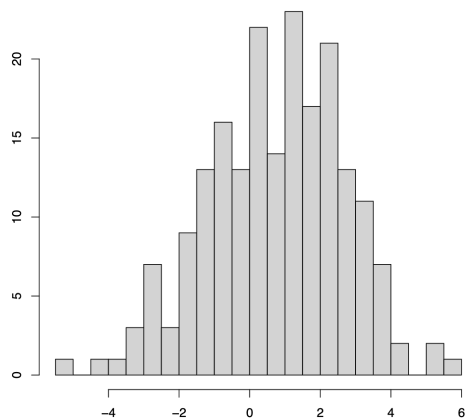


Figure 2: Normal distribution with mean $\mu = 1$ and $\sigma = 2$.

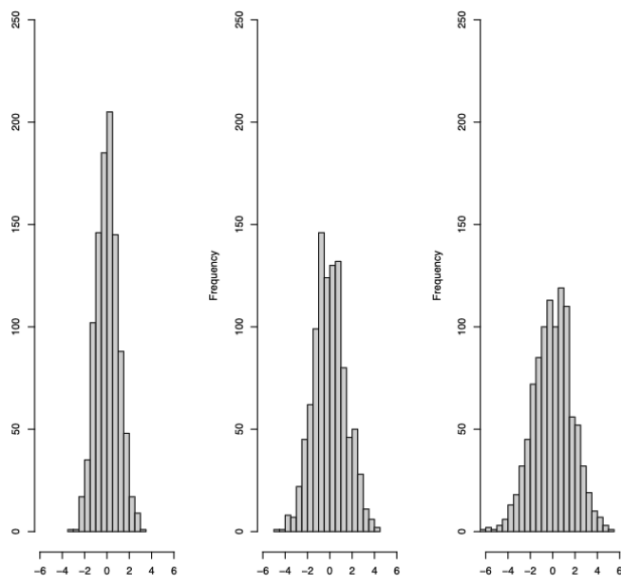


Figure 3: Different standard deviations result in differently shaped distributions.